

GARV ARORA

Robotics Software Engineer

Vellore, India | +91 88008-12254 | garvarora0205@gmail.com | linkedin.com/in/gaminization | github.com/gaminization

SUMMARY

Final-year Computer Science undergraduate (IoT) at VIT with 2 years building ROS2 navigation stacks, CUDA-accelerated CV pipelines, and ARM embedded systems. Ranked 12th globally at IRC 2025 competing against 100+ international teams. Targeting entry-level software engineering or robotics roles.

EDUCATION

Vellore Institute of Technology | Vellore, India

B.Tech – Computer Science & Engineering (Specialization: IoT)

CGPA: 8.37 / 10

Aug 2023 – Jul 2027

Gems International School | Gurugram, India

Central Board of Secondary Education (CBSE)

Class 10: 89.3% | Class 12: 76.4%

Nov 2016 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS

Robotics & Computer Vision: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, PID Control, EKF, OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK

ML & Frameworks: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, Flask, React, Node.js

Infrastructure: Linux, Docker, Git, GitLab CI/CD, AWS, CUDA, PyQt, MongoDB

WORK EXPERIENCE

Team Vyadh – Martian Rover Team

Autonomous Systems Developer

Apr 2024 – Present

Vellore, India

- Wrote a full ROS2 autonomous navigation stack integrating Nav2, RTAB-Map Visual SLAM, and Intel RealSense D455, producing real-time 3D elevation maps for path planning and obstacle avoidance across unstructured terrain.
- Reduced CV pipeline frame latency by ~60% and raised throughput 4x by combining CUDA acceleration, YOLOv8 detection, ArUco/AprilTag recognition, and Linux V4L2 buffer tuning across 4 camera streams.
- Flashed Micro-ROS firmware to 3 ESP32 nodes for distributed sensor acquisition and actuator control; fused IMU + encoder data with an Extended Kalman Filter to cut odometry drift and enable PID-based closed-loop mobility.

Wissen Baum Engineering Solutions

Software Automation Intern

May 2025 – Jul 2025

Pune, India

- Wrote a BDD test suite (Python, Behave, pytest) covering 40+ end-to-end scenarios; parallelized execution across 3 GitLab CI/CD stages with Playwright and Cypress, cutting pipeline run time by ~35%.
- Replaced manual QA data checks with a pandas/NumPy validation script that flagged schema errors across 10,000+ rows per run; added pre-commit Git hooks to block non-compliant pushes.

PROJECTS

HayaiOS | C, ARM Assembly

2026

- Wrote a bare-metal RTOS on ARM Cortex-M4 from scratch: preemptive round-robin scheduler with 1 ms tick, context switching in <50 assembly instructions, mutex/semaphore primitives, and a HAL isolating all register-level I/O.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Fused RealSense D455 depth frames with IMU data via complementary filter to stabilize KinectFusion TSDF reconstruction during fast camera motion, reducing tracking failure rate on a 6-DOF test trajectory.

PDF Summarizer | Python, PyMuPDF, HuggingFace Transformers, Docker

2025

- Parsed and ranked sections across multiple PDFs with PyMuPDF; queried a HuggingFace summarization model via RAG-style retrieval and shipped the pipeline as a Docker image for one-command deployment.

Catters – Endless Runner | Unity, C#

2024

- Shipped an Android game to device: procedural jungle generation, touch input, collision system, and score-based difficulty curve built in Unity within a 3-week solo sprint.

CERTIFICATIONS & ACHIEVEMENTS

IRC 2025: Ranked 12th globally out of 100+ international teams | **IRDC 2025:** Top 5 Finalist | **Oracle Cloud Infrastructure 2025:** Generative AI Professional, Data Science Professional